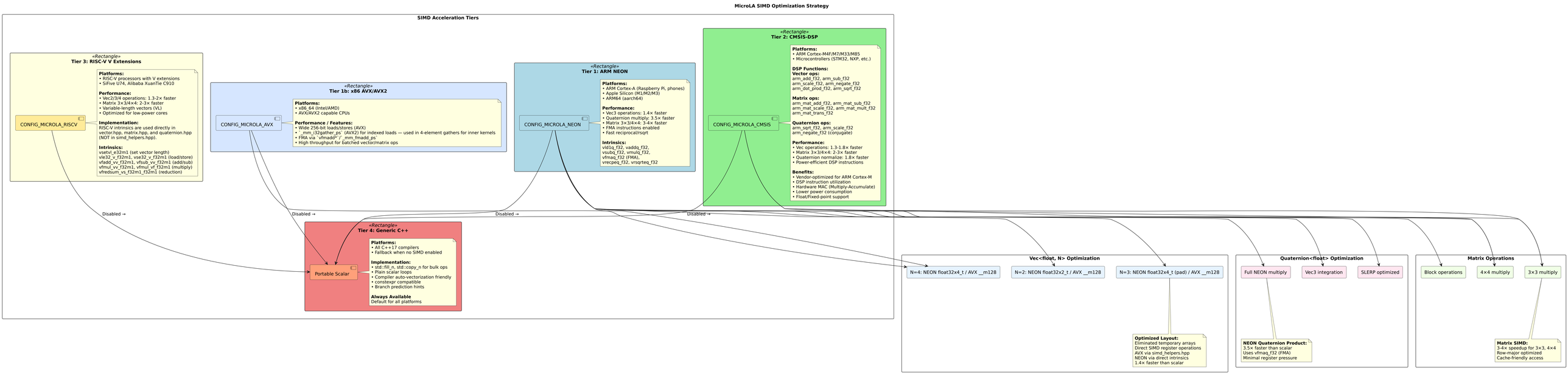
MicroLA SIMD Optimization Strategy
SIMD Acceleration Tiers
«Rectangle»
Tier 3: RISC-V V Extensions
Platforms:
• RISC-V processors with V extensions
• SiFive U74, Alibaba XuanTie C910
Performance:
• Vec2/3/4 operations: 1.3-2× faster
• Matrix 3×3/4×4: 2-3× faster
• Variable-length vectors (VL)
• Optimized for low-power cores
Implementation:
RISC-V intrinsics are used directly in vector.hpp, matrix.hpp, and quaternion.hpp (NOT in simd_helpers.hpp).
Intrinsics:
vsetvl_e32m1 (set vector length)
vle32_v_f32m1, vse32_v_f32m1 (load/store)
vfadd_vv_f32m1, vfsub_vv_f32m1 (add/sub)
vfmul_vv_f32m1, vfmul_vf_f32m1 (multiply)
vfredsum_vs_f32m1_f32m1 (reduction)
CONFIG_MICROLA_RISCV
«Rectangle»
Tier 1b: x86 AVX/AVX2
Platforms:
• x86_64 (Intel/AMD)
• AVX/AVX2 capable CPUs
Performance / Features:
• Wide 256-bit loads/stores (AVX)
• `_mm_i32gather_ps` (AVX2) for indexed loads — used in 4-element gathers for inner kernels
• FMA via `vfmadd*` / `_mm_fmadd_ps`
• High throughput for batched vector/matrix ops
CONFIG_MICROLA_AVX
«Rectangle»
Tier 1: ARM NEON
Platforms:
• ARM Cortex-A (Raspberry Pi, phones)
• Apple Silicon (M1/M2/M3)
• ARM64 (aarch64)
Performance:
• Vec3 operations: 1.4× faster
• Quaternion multiply: 3.5× faster
• Matrix 3×3/4×4: 3-4× faster
• FMA instructions enabled
• Fast reciprocal/rsqrt
Intrinsics:
vld1q_f32, vaddq_f32,
vsubq_f32, vmulq_f32,
vfmaq_f32 (FMA),
vrecpeq_f32, vrsqrteq_f32
CONFIG_MICROLA_NEON
«Rectangle»
Tier 2: CMSIS-DSP
Platforms:
• ARM Cortex-M4F/M7/M33/M85
• Microcontrollers (STM32, NXP, etc.)
DSP Functions:
Vector ops:
arm_add_f32, arm_sub_f32
arm_scale_f32, arm_negate_f32
arm_dot_prod_f32, arm_sqrt_f32
Matrix ops:
arm_mat_add_f32, arm_mat_sub_f32
arm_mat_scale_f32, arm_mat_mult_f32
arm_mat_trans_f32
Quaternion ops:
arm_sqrt_f32, arm_scale_f32
arm_negate_f32 (conjugate)
Performance:
• Vec operations: 1.3-1.8× faster
• Matrix 3×3/4×4: 2-3× faster
• Quaternion normalize: 1.8× faster
• Power-efficient DSP instructions
Benefits:
• Vendor-optimized for ARM Cortex-M
• DSP instruction utilization
• Hardware MAC (Multiply-Accumulate)
• Lower power consumption
• Float/Fixed-point support
CONFIG_MICROLA_CMSIS
«Rectangle»
Tier 4: Generic C++
Platforms:
• All C++17 compilers
• Fallback when no SIMD enabled
Implementation:
• std::fill_n, std::copy_n for bulk ops
• Plain scalar loops
• Compiler auto-vectorization friendly
• constexpr compatible
• Branch prediction hints
Always Available
Default for all platforms
Portable Scalar
Disabled →
Disabled →
Disabled →
Disabled →
Vec<float, N> Optimization
N=4: NEON float32x4_t / AVX __m128
N=2: NEON float32x2_t / AVX __m128
N=3: NEON float32x4_t (pad) / AVX __m128
Optimized Layout:
Eliminated temporary arrays
Direct SIMD register operations
AVX via simd_helpers.hpp
NEON via direct intrinsics
1.4× faster than scalar
Quaternion<float> Optimization
Full NEON multiply
Vec3 integration
SLERP optimized
NEON Quaternion Product:
3.5× faster than scalar
Uses vfmaq_f32 (FMA)
Minimal register pressure
Matrix Operations
Block operations
4×4 multiply
3×3 multiply
Matrix SIMD:
3-4× speedup for 3×3, 4×4
Row-major optimized
Cache-friendly access